

The title

true

true

Introduction

The shape bias is the tendency of young children to generalize new words based on their shape (Landau et al., 1988). The shape bias is the tendency to generalize a novel noun to new category members based on shape similarity rather than other features such as color or material (Landau, Smith, & Jones, 1988).

One reason for researchers’ long-standing interest in the shape bias relates to its hypothesized role in early word learning. This bias has been extensively studied in English-speaking children in the US, where it emerges around 18-24 months of age and is thought to be a key mechanism for rapid vocabulary growth during the so-called “vocabulary spurt” (Gershkoff-Stowe & Smith, 2004; Smith, 2000).

It is reliably observed in English-speaking children but reported to vary across languages (Abdelrahim & Frank, 2024); e.g., it is argued to be lower in East Asian languages such as Korean (Gathercole & Min, 1997).

The shape bias is proposed to be a statistical generalization of early vocabulary: in English, solid objects are dominant in early lexicon, are referred to by count nouns, and are organized by shape (Samuelson & Smith, 1999). The emergence of the shape bias has been linked to the distributional properties of early-learned nouns in English, which tend to refer to solid objects that are countable and organized by shape (Samuelson & Smith, 1999). This was also supported by computational models showing that exposure to a lexicon with similar statistical properties can lead to the emergence of a shape bias in simulated learners (Colunga & Smith, 2005; Perry & Samuelson, 2011). A study utilizing bayesian models further demonstrated that the shape bias can emerge from the statistical structure of the lexicon, specifically the correlations between object properties (solidity, shape) and linguistic properties (count-mass syntax) (Kemp et al., 2007).

Recent work on large language models

However, the extent to which the shape bias and its underlying lexical statistics are universal across languages remains unclear. Some studies have suggested that the shape bias may be less pronounced or absent in languages with different noun classification systems or morphological structures (e.g., Chinese, Japanese) (Imai et al., 2008; Yoshida & Smith, 2005). Other studies have found evidence for a shape bias in non-English-speaking children, but with variations in its strength and developmental trajectory (e.g., Spanish, German) (Casasola & Bhagwat, 2007; Graham et al., 2004). To address this gap, we conducted a cross-linguistic study to examine the lexical statistics of early-learned nouns in 16 languages from diverse linguistic families and regions. We collected adult judgments on the same set of nouns across three dimensions: solidity (solid vs. non-solid), count-mass syntax (count vs. mass), and shape-based categorization (shape, color, material, none). We then analyzed the distributions of nouns across these dimensions and their correspondence within and across languages. We also tested for differences in the distribution of shape-based categorization responses across languages using a permutation test.

We tested three predictions of a statistical account: cross-linguistic variation in lexical statistics should predict the reported cross-cultural variation in the shape bias; the shape bias associated with individual words should predict when they are learned (their age of acquisition; AoA); and cross-cultural judgments of lexical statistics should predict the magnitude of children’s shape bias.

Study 1A:

In this study we first aim to replicate Samuelson et al 1999 where they asked only 11 adults to make judgements on the first 300 words learned by children in the US English on three dimensions: solidity, count-mass syntax, and shape-based categorization. We extend this by collecting ratings on the same set of words from a larger sample of English-speaking adults recruited online from the Prolific platform. Most importantly, we also collected ratings on words from other Languages. ## Methods

In Study 1, we collect judgment ratings on the first 300 words learned from the MCDI US English across the same three dimensions of solidity, count, and mass syntax, and shape-based categorization. The study, however, collected these ratings on the first 300 words learned by children in sixteen languages including English.

Participants

377 English-speaking adults recruited online from the Prolific platform (mean age = 41, sd = 14) were asked to rate words in three dimensions: solidity (solid, non-solid, unclear), count and mass noun syntax (count, mass, unclear), and organizing feature (shape, color, material, none of these).

Material

Three hundred nouns were sourced from MCDI data on the WordBank repository of 16 languages that belonged to seven linguistic families: (LIST OF LANGUAGES). The nouns were chosen based on the earliest age of acquisition in the dataset, up to a maximum of 300 nouns (CITE). Because we only recruited English-speaking participants, Nouns were presented in English and mapped across languages via conceptual mappings (“unilemmas”, Frank et al., 2021). A “unilemma” is the concept behind the noun expressed in the English language (“dog” and “dog in german” both map to the same underlying concept of “dog”, which is used as the unilemma in this case).

The final sample consisted of a total of 23355 nouns from all languages, with $N = \sim 664$ after removing duplicates “uni-lemmas shared between languages”. (Figure to show how much overlap between languages). With the Semantic categories breakdown of the words in each language attached in the supplementary materials. Most words belonged to the categories of food/drinks, clothing, and animals, with a few words referring to vehicles, toys, body parts, household items, and outside things.

Procedure

The experiment started with familiarization trials in which participants were shown examples of count and mass nouns, solid objects, and non-solid objects. They were also familiarized with what a category-organizing or characterizing feature means, as in Table [2]. Then the test trials followed the same wording and structure. Every dimension (solidity, count-mass syntax, and category organizing feature) constituted a block, and every participant rated 20 words per block, so there were 60 test trials in total. Familiarization trials, test blocks, and words within a block were all randomized across participants. No information about the semantic category of words was given to participants, nor were the words grouped by their semantic category. Participants were only allowed to choose one feature in all questions.

Data analysis

We calculated the proportion of responses for each feature per word, separately for each language. For example, if a word was rated by 10 participants, and 7 of them rated it as solid, the proportion of solid responses for that word would be 0.7. We then conducted a permutation test to compare the observed

density of shape-based categorization responses against a null distribution generated by permuting language labels. The permutation test involved the following steps: 1. Isolate the data for shape-based categorization responses. 2. Decide on a grid of proportion-values between 0 and 1. 3. Compute the observed density of shape-based categorization responses for each language using kernel density estimation. 4. Generate a null distribution by randomly permuting language labels and recomputing the density for each permutation. 5. Compare the observed density against the null distribution to assess whether there are significant differences in shape-based categorization across languages.

Additionally, and similar to Samuelson et al 1999, we categorized the words into three categories: solid, count noun, and shape-based categorization using a cutoff of 80% i.e. a word that is rated by at least 80% of the participants to be a solid is categorized as solid. We then visualized the overlap between these categories using Venn diagrams for each language.

lastly, we assessed the correspondence between the three dimensions using Pearson correlation analysis. We also calculated inter-rater agreement using the word-level consensus approach, which involves calculating the mode of the ratings for each word and then computing the percentage of raters who agreed with the mode. This gives us a measure of how consistently participants rated each word across the three dimensions.

Results

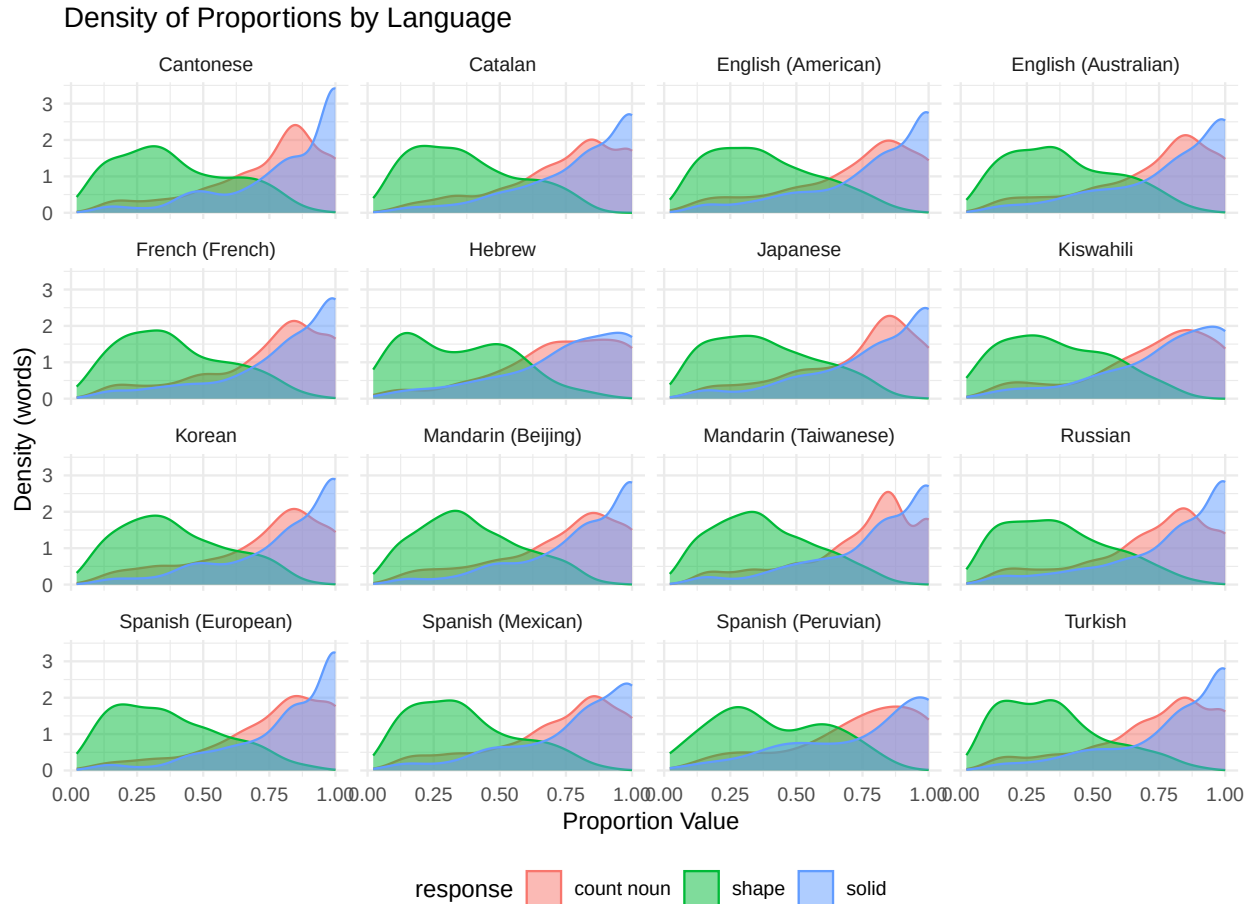


Figure 1: This is a figure caption.

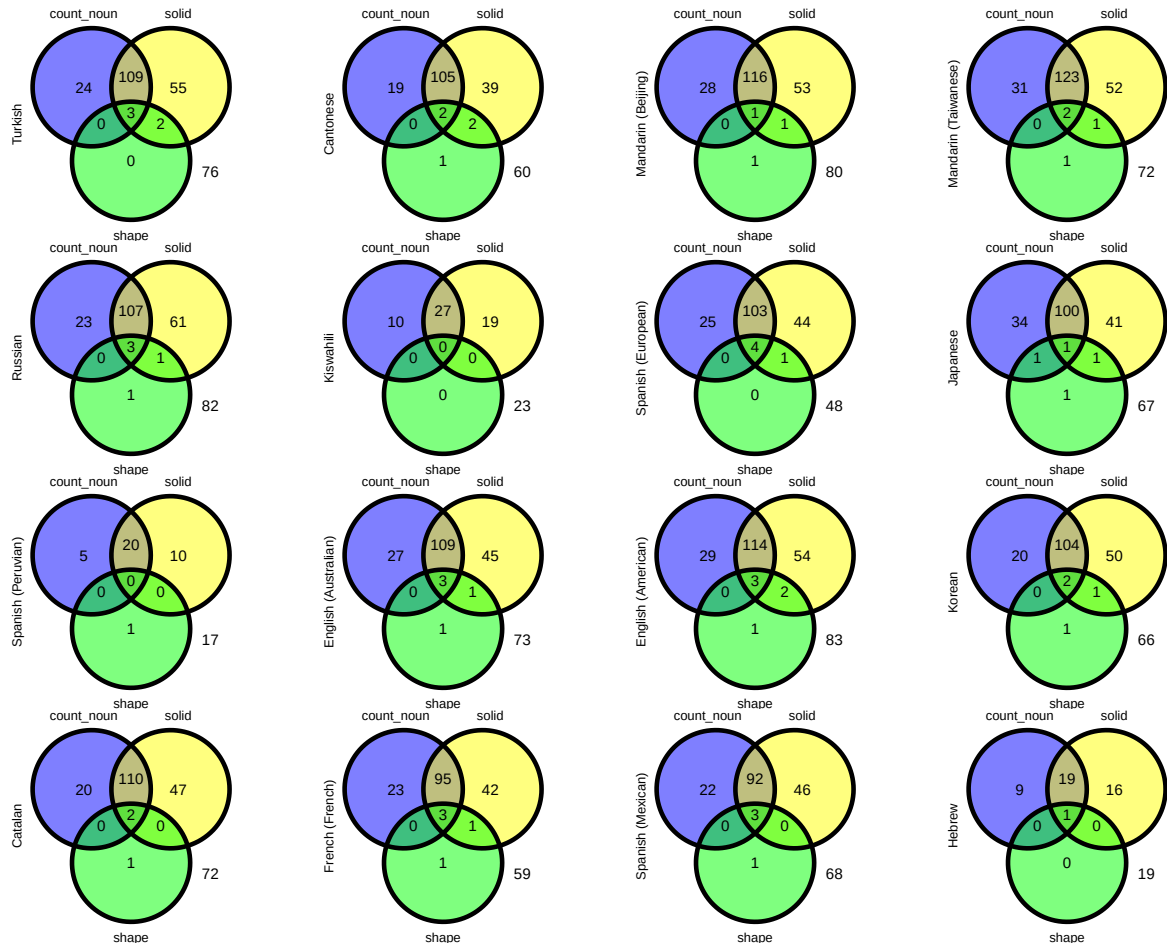
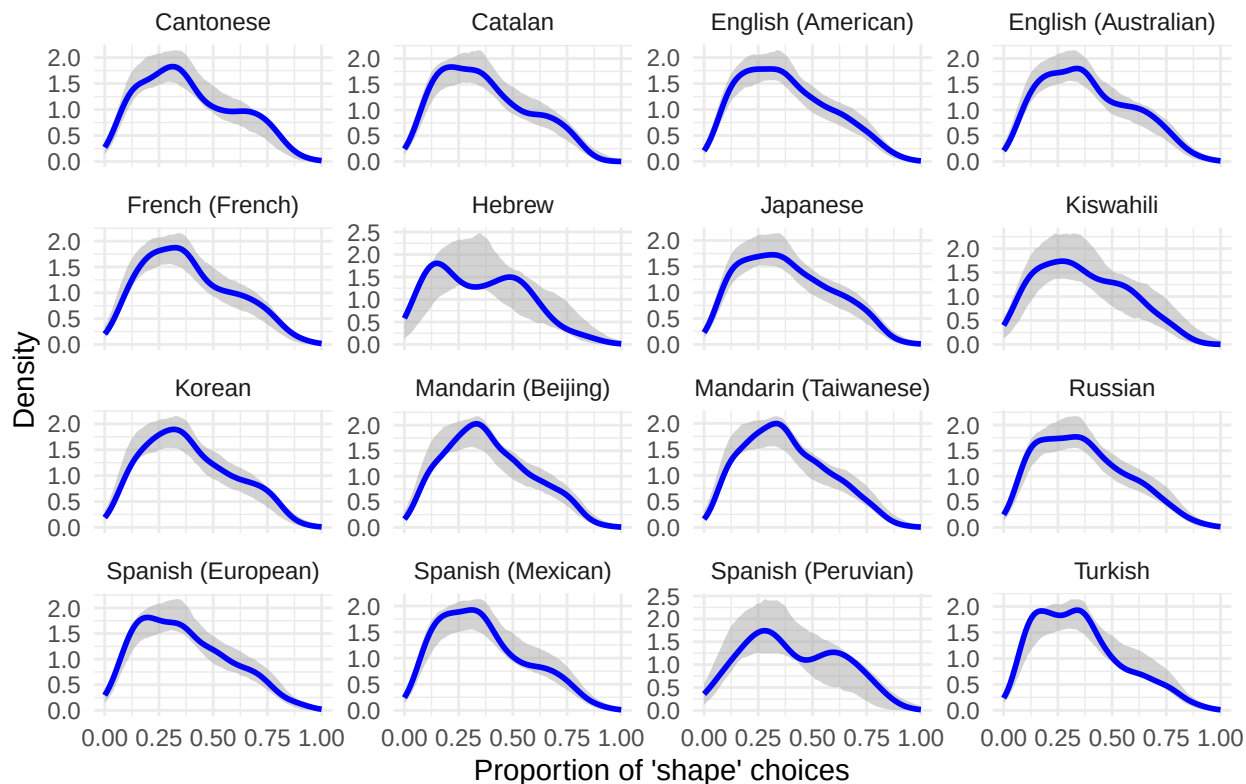


Figure 2: Venn diagram showing the overlap between nouns categorized as solid, count nouns, and organized by shape in English (Study 1A). A cutoff of 80% was used to categorize words, meaning that a word rated by at least 80% of participants as solid is categorized as solid.

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Observed 'shape' density vs. Null Envelope by Language

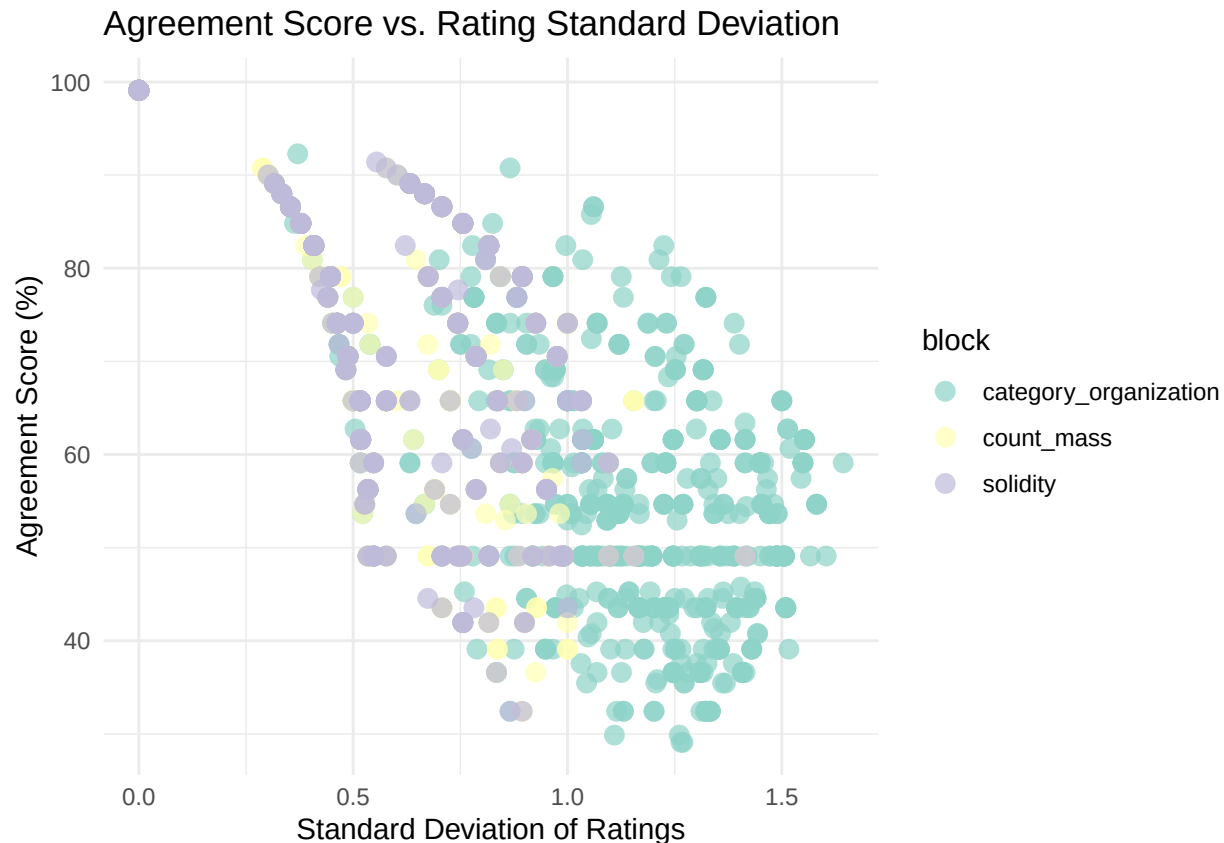


We collected a total of 20,130 ratings across 377 participants for 664 words in three blocks (solidity, count-mass syntax, and category-organizing feature). After removing responses with no answer, we retained 19,635 ratings. Each word received an average of between 8 and 10 ratings per block. Included in the supplementary materials are the demographic details of the participants (Table 1) and the distribution of ratings across words and blocks (Figure 1).

Across languages, ratings were very consistent across languages, suggesting limited variation in lexical statistics as can be seen in figure @ref{fig:correspondenceeng1}. Nouns were predominantly rated as solid ($M = 0.72$, $SD = 0.32$) and count ($M = 0.68$, $SD = 0.29$), but were less reliably judged as being organized by shape ($M = 0.31$, $SD = 0.24$). Pearson correlation analysis revealed a moderate positive correlation between solidity and count syntax ($r = 0.62$, $p < 0.001$), a weak positive correlation between count syntax and shape-based categorization ($r = 0.45$, $p < 0.001$), and a weaker positive correlation between solidity and shape-based categorization ($r = 0.38$, $p < 0.001$). The venn diagram in figure ?? shows categories of words as solid, count noun, and shape_characterized by using a cutoff of 80% i.e. a word that is rated by at least 80% of the participants to be a solid is categorized as solid. The venn diagram shows that a large number of nouns were consistently rated as solid and count nouns, with more overlap between these two dimensions. However, the minority of words rated as referring to objects organized by shape. This set of words rated to be organized by shape is significantly smaller than the set reported in Samuelson 1999.

To test for distributional differences in shape responses across languages, we conducted a permutation test (see Methods). The results showed that the observed density curves for each language fell within the null distribution generated by permuting language labels, indicating no significant differences in lexical statistics across languages ??.

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Discussion

The emergence of the shape bias as a word learning strategy in English speaking kids in the US, is thought to be a product of the distribution of early learned count nouns that correspond to solid objects which happen to have similar shapes. Their attention is thus tuned to shape as the relevant feature for categorizing and correctly extending nouns to new members of the referent category. Even the fewest set of noun-referents that follow this distribution and its correspondence is sufficient to trigger the shape bias in toddlers. Following from this, we expected to find significant cross-linguistic variation in the distribution of nouns across these three dimensions, which would then predict differences in the emergence of the shape bias across languages. However, our results showed that lexical statistics were remarkably consistent across languages. Across the three dimensions, distributions of all languages look the same and are not significantly different from the null distribution using a permutation test. Correspondence between the ontological status, syntactic frame, and shape organization was also very weak.

However, maybe the question about the organizing feature was not clear, not intuitive for a naive participants, and instructions were ambiguous, which is why solidity and countability correspond to each other but less with shape ratings. The weak shaped based choices and large SD, along with the shape-solidity correlation ($r = 0.38$) might partly reflect measurement error. We test the reliability of these measure in study 1B.

Study 1B

The idea of the organizing feature of an object category is not an intuitive one, and is not something that people actively think about when recognizing category members i.e. when they are searching for apples, they don't break down the category into features of apple shape, red color, and leaf material and then decide

which feature is the most diagnosing of apples. It is possible that the instructions were not clear enough for participants to understand what was being asked. To test the reliability of this measure, we collected a second sample of ratings on the same set of words, with improved instructions and examples.

Participants

377 English-speaking adults recruited online from the Prolific platform (mean age = 41, sd = 14) were asked to rate words in three dimensions: solidity (solid, non-solid, unclear), count and mass noun syntax (count, mass, unclear), and organizing feature (shape, color, material, none of these).

Material

The same set of words used in Study 1A. However, only the category-organizing feature block was included in this study.

Procedure

The experiment started with familiarization trials of what a category-organizing or characterizing feature means. However, this time the instructions were improved with more examples, adding picture depictions, training questions with feedback and clarifications involving the four feature answers so not to bias subjects into a specific answer as in Table [2]. The rest of the study was similar to Study 1A.

Data analysis

Similar to study 1A, we calculated the proportion of responses for each feature per word, and then conducted a permutation test to compare the observed density of shape-based categorization responses against a null distribution generated by permuting language labels.

Results

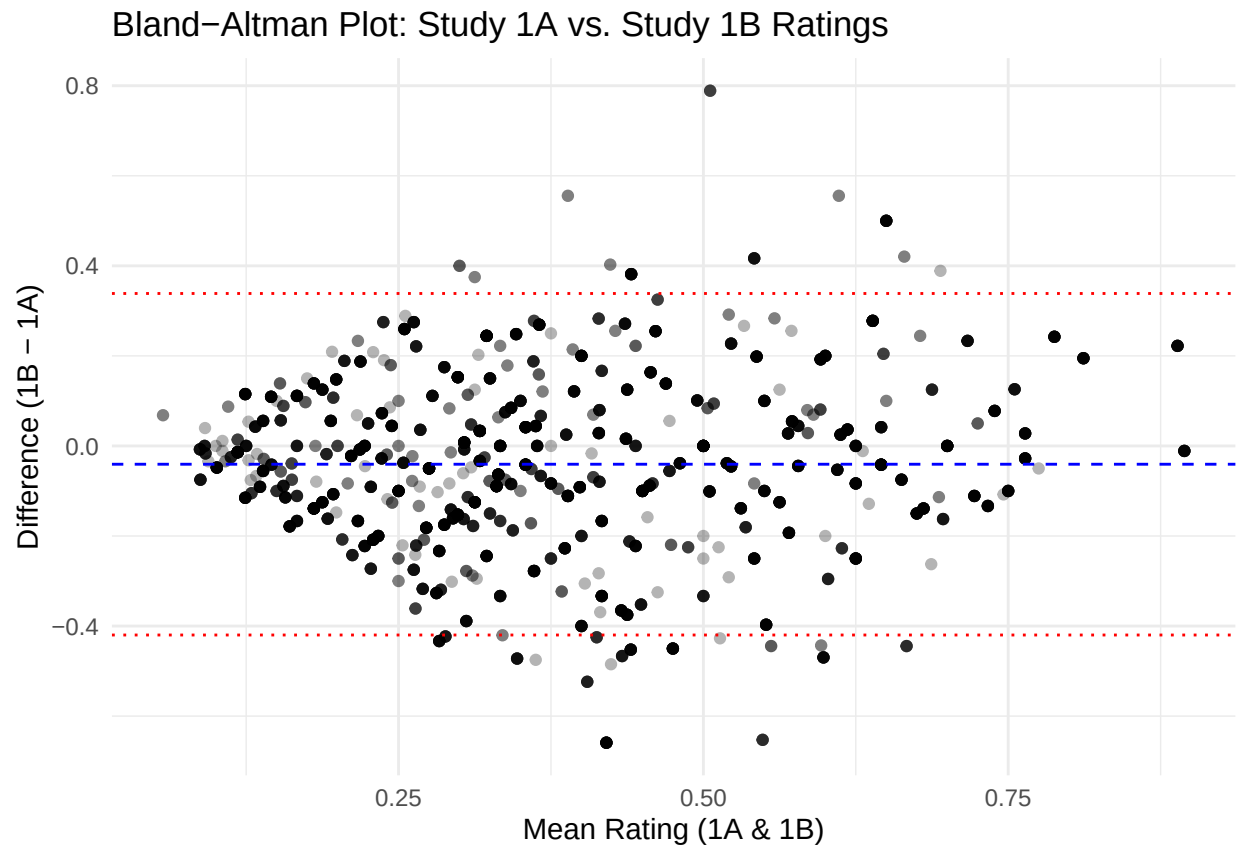
```
## [1] "Intraclass Correlation Coefficients:"

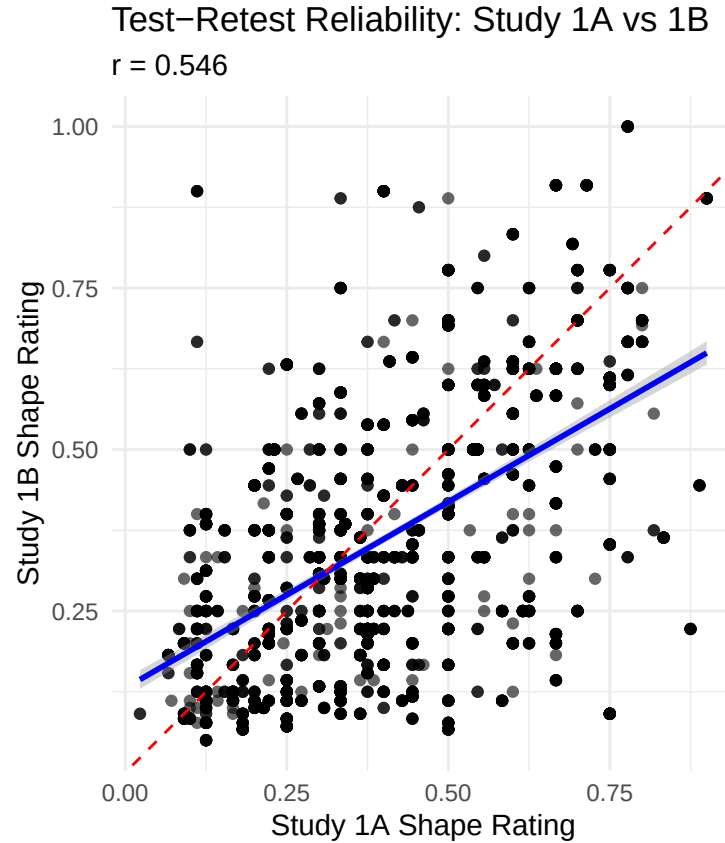
## Call: ICC(x = icc_data)
##
## Intraclass correlation coefficients
##
```

	type	ICC	F	df1	df2	p	lower bound
## Single_raters_absolute	ICC1	0.53	3.3	2621	2622	7.3e-190	0.50
## Single_random_raters	ICC2	0.53	3.4	2621	2621	6.5e-203	0.49
## Single_fixed_raters	ICC3	0.54	3.4	2621	2621	6.5e-203	0.52
## Average_raters_absolute	ICC1k	0.69	3.3	2621	2622	7.3e-190	0.67
## Average_random_raters	ICC2k	0.70	3.4	2621	2621	6.5e-203	0.66
## Average_fixed_raters	ICC3k	0.71	3.4	2621	2621	6.5e-203	0.68

```
##
## upper bound
## Single_raters_absolute      0.56
## Single_random_raters      0.57
## Single_fixed_raters       0.57
## Average_raters_absolute    0.72
## Average_random_raters     0.73
## Average_fixed_raters      0.73
```

```
##  
## Number of subjects = 2622      Number of Judges = 2  
## See the help file for a discussion of the other 4 McGraw and Wong estimates,
```





Similarly, ratings of the organizing feature in study 1B were consistent across all languages and didn't differ significantly from the null distribution (Figure ??). With a mean of () shape ratings and high standard deviation of (). The distribution of shape-based categorization responses was generally similar to that observed in Study 1A. We compared shape ratings between Study 1A and Study 1B using both mixed-effects modeling and paired comparisons. Ratings in Study 1B were slightly but significantly lower than in Study 1A ("B" = -0.04, $p < .001$), with a small effect size ($d^2 = 0.04$, $d = 0.18$).

A Bland-Altman analysis indicated that, while (mean agreement was high?) the ICC reliability estimate was very precise (95% CI [0.52, 0.57]), individual word differences ranged from -0.42 to +0.34, reflecting some variability at the item level.

Reliability analysis

we only assessed the reliability of the organizing feature, not the solid or countable questions. We assessed the consistency of shape ratings between Study 1A and Study 1B using both Pearson correlation and Intraclass Correlation Coefficients (ICC). The test-retest correlation was moderate ($r = 0.546$). ICC estimates indicated good agreement between studies (ICC ~ 0.70 for average ratings), suggesting that despite some individual variability, the overall rating patterns are reliably replicable across studies. in how participants rated the organizing feature of nouns across the two samples. This suggests that participants understood the concept of an organizing feature and were able to reliably apply it to a wide range of nouns. The correlation for the 'none of these' option was ($r = 0.58$, $p < 0.001$), suggesting that this option may have been used more consistently across studies, possibly reflecting a clearer understanding of when none of the provided features applied.

Discussion

The Systematic Difference in the significant decrease in Study 1B (better instructions led to LOWER shape ratings) suggests: Possible Interpretation 1: Study 1A instructions were ambiguous and led to inflated shape ratings Possible Interpretation 2: Study 1B instructions were overly restrictive Possible Interpretation 3: The construct itself is inherently difficult to measure reliably The moderate test-retest reliability ($r = 0.55$) of the organizing feature measure represents a significant limitation. This finding suggests that categorizing objects by their organizing feature may be inherently difficult for naive participants, even with improved instructions. This is consistent even with the results from the ‘none of these’ option, which showed a slightly higher reliability ($r = 0.58$), indicating that participants were more consistent when indicating that none of the provided features applied to a noun. It is also consistent with Samuleson et al 1999 results with shape rating = , $SD=$, showing that adults are not consistent in their judgments of shape similarity. Future research should explore alternative approaches to measuring shape-based organization, such as forced-choice paradigms or implicit measures.

Given the moderate reliability of the organizing feature measure, we prioritized Study 1B ratings in subsequent analyses as they were collected with improved instructions and training. For robustness, we also conducted sensitivity analyses excluding words with the largest between-study differences.

Age of acquisition (AoA)

why AoA?

Focus on strongest effects only Position findings as exploratory

Method

We obtained AoA ratings for the nouns in our dataset from the MRC Psycholinguistic Database (Coltheart, 1981) and the Kuperman et al. (2012) database. For words present in both databases, we averaged the AoA ratings to obtain a single value per word. AoA ratings were available for 500 of the 664 nouns in our dataset, with a mean AoA of 6.5 years ($SD = 2.1$ years).

Data Analysis

Results

We examined the relationship between AoA and shape-based categorization ratings using Pearson correlation analysis. The analysis revealed a weak negative correlation between AoA and shape ratings ($r = -0.15$, $p < 0.01$), suggesting that words learned earlier in life tend to be rated slightly higher on shape-based categorization. However, this relationship was not strong, indicating that other factors likely play a more significant role in determining shape-based categorization ratings.

Qualitative Examples: a few examples of words that are high on solid rating but perhaps lower on shape or concreteness (or have other characteristics) and have later AoAs. This makes the “residual unique effect” argument more tangible. (e.g., “Examples of words rated high on solidity but acquired later included X, Y, Z, which are typically [describe their characteristics, e.g., complex, less manipulable, etc.]”).

Study 2:

Methods

Participants

Material

Procedure

Data analysis

Results

Discussion